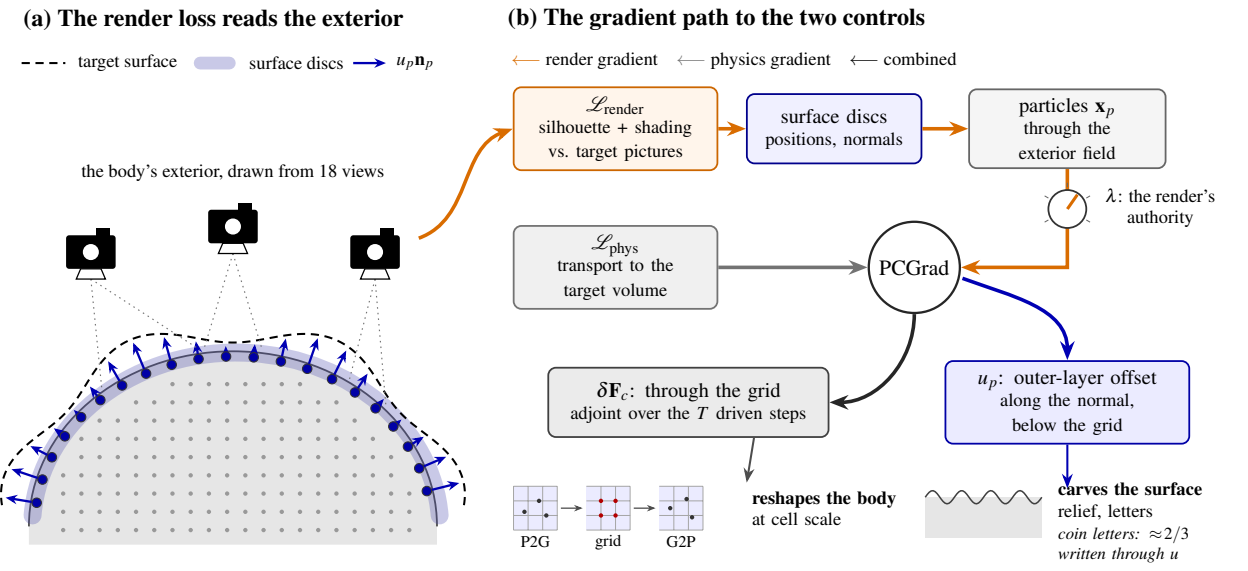

(a) The render loss reads the exterior
target surface
surface discs
$u_p\mathbf{n}_p$
the body's exterior, drawn from 18 views
(b) The gradient path to the two controls
render gradient
physics gradient
combined
$\mathscr{L}_{\text{render}}$
silhouette + shading
vs. target pictures
surface discs
positions, normals
particles $\mathbf{x}_p$
through the
exterior field
λ: the render's
authority
$\mathscr{L}_{\text{phys}}$
transport to the
target volume
PCGrad
$\delta\mathbf{F}_c$: through the grid
adjoint over the T driven steps
u_p: outer-layer offset
along the normal,
below the grid
P2G
grid
G2P
reshapes the body
at cell scale
carves the surface
relief, letters
coin letters: ≈2/3
written through u